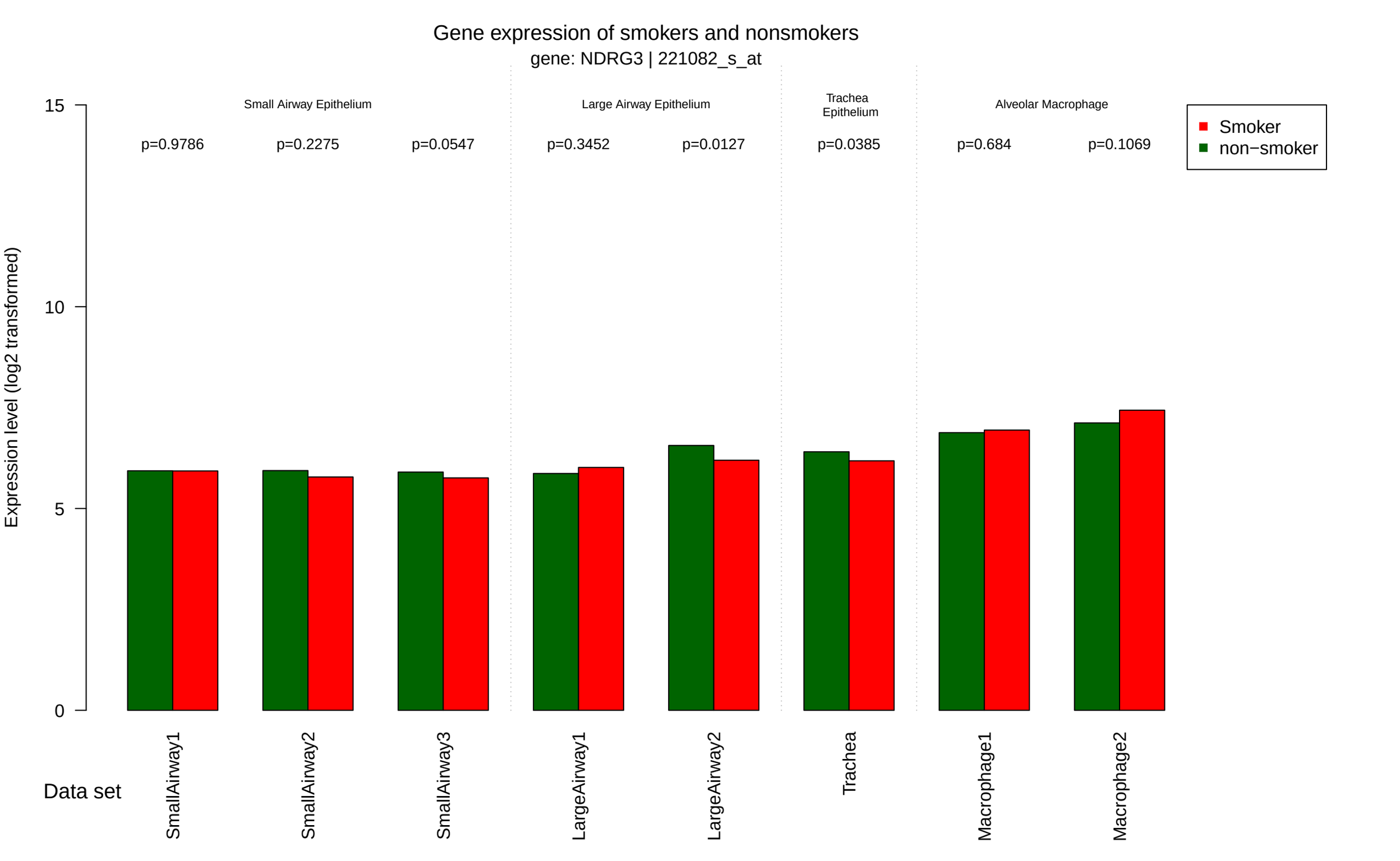
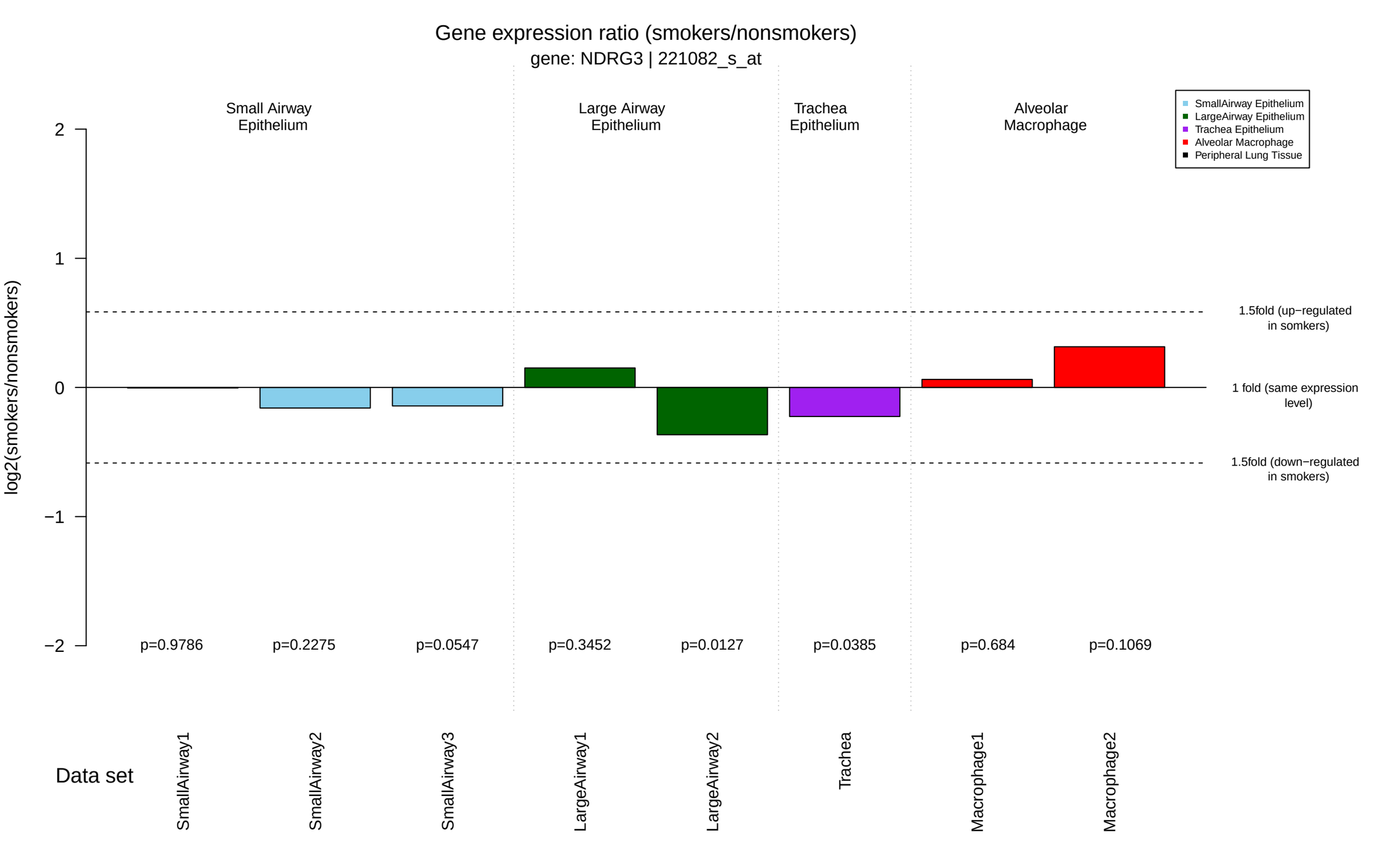
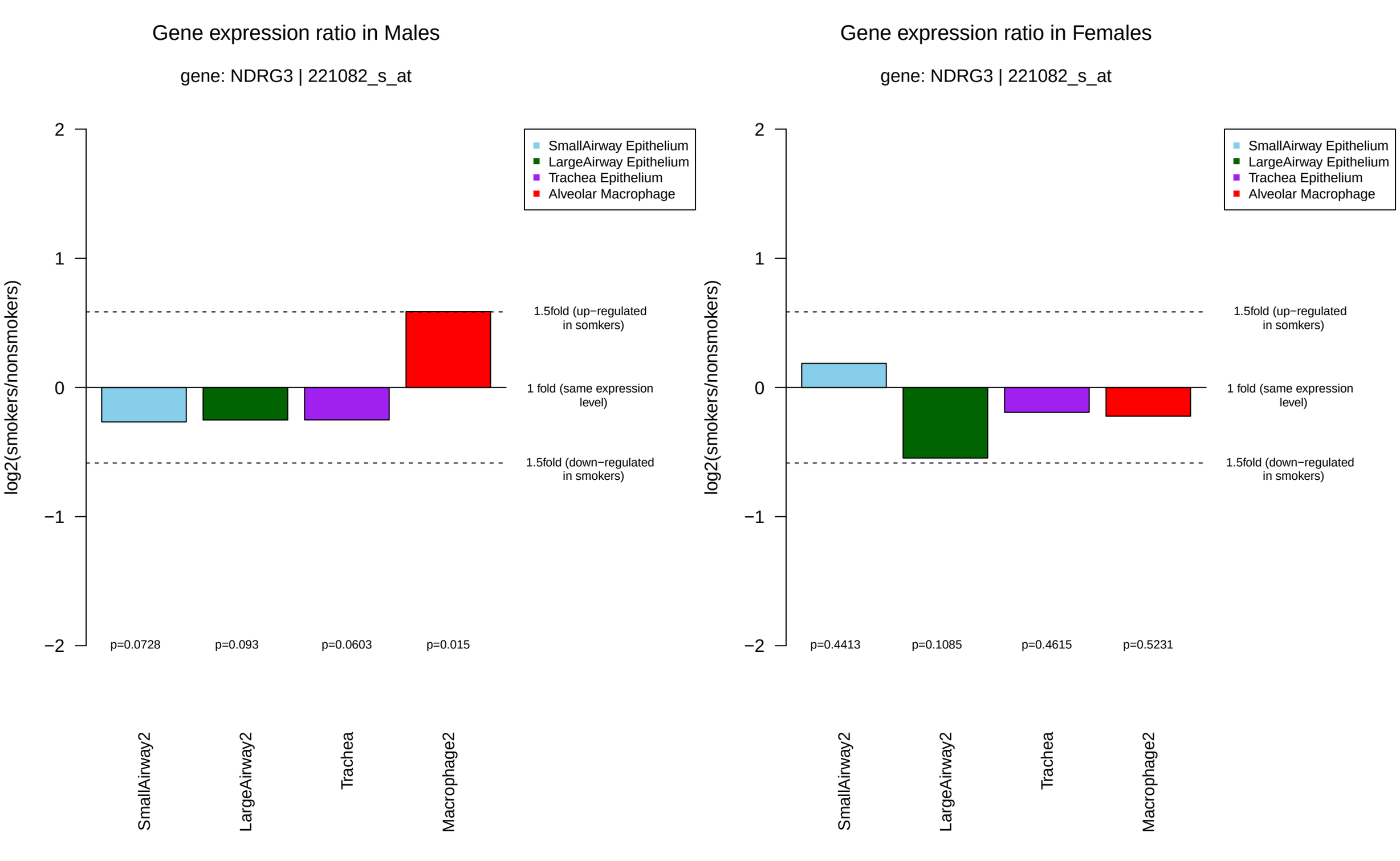
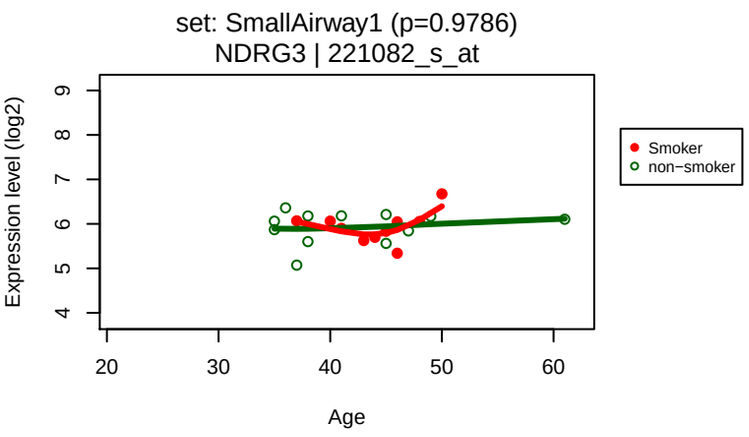
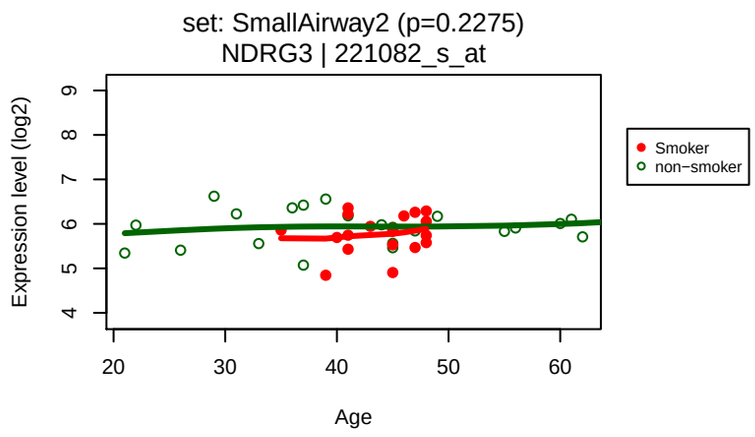
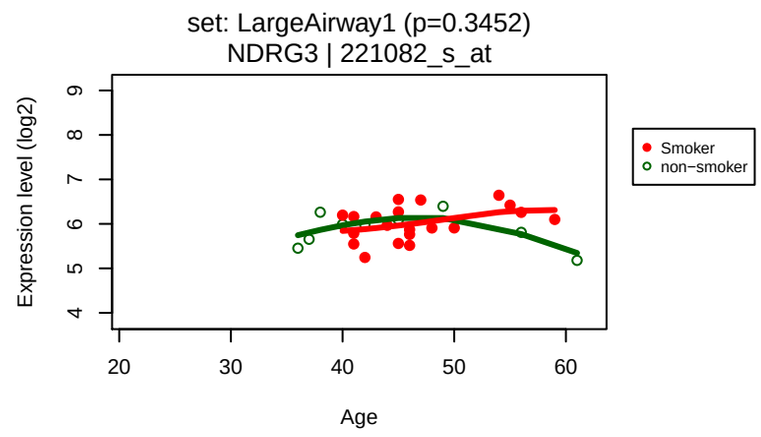

Gene expression of smokers and nonsmokers
gene: NDRG3 | 221082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9786
p=0.2275
p=0.0547
p=0.3452
p=0.0127
p=0.0385
p=0.684
p=0.1069
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

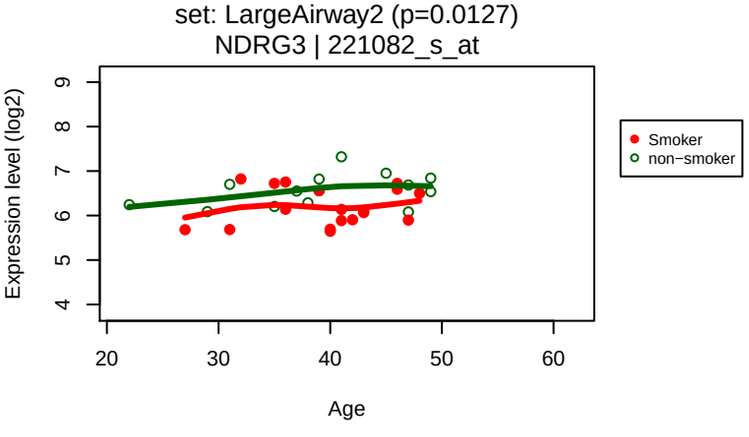

Gene expression ratio (smokers/nonsmokers)
gene: NDRG3 | 221082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9786
p=0.2275
p=0.0547
p=0.3452
p=0.0127
p=0.0385
p=0.684
p=0.1069
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

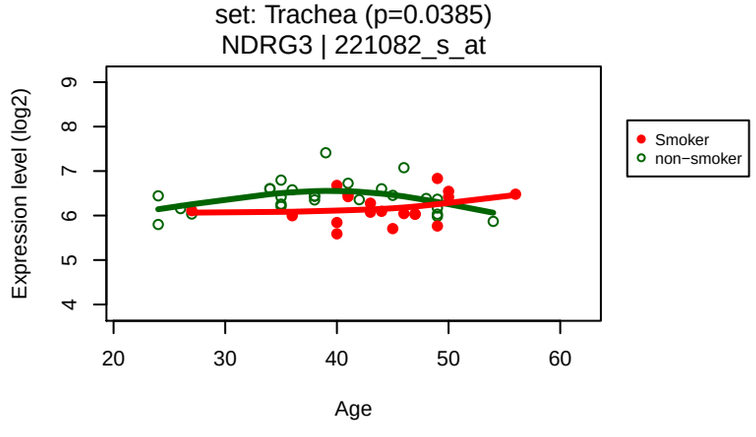

Gene expression ratio in Males
gene: NDRG3 | 221082_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0728
p=0.093
p=0.0603
p=0.015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NDRG3 | 221082_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4413
p=0.1085
p=0.4615
p=0.5231
SmallAirway2
LargeAirway2
Trachea
Macrophage2

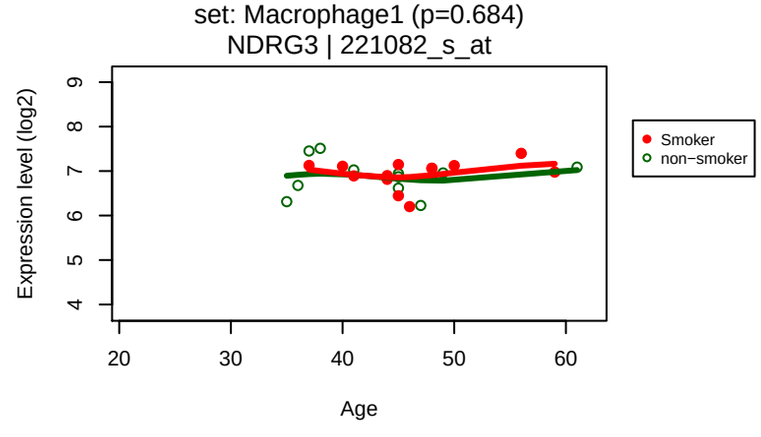

set: SmallAirway1 (p=0.9786)
NDRG3 | 221082_s_at
Expression level (log2)
Age
Smoker
non−smoker

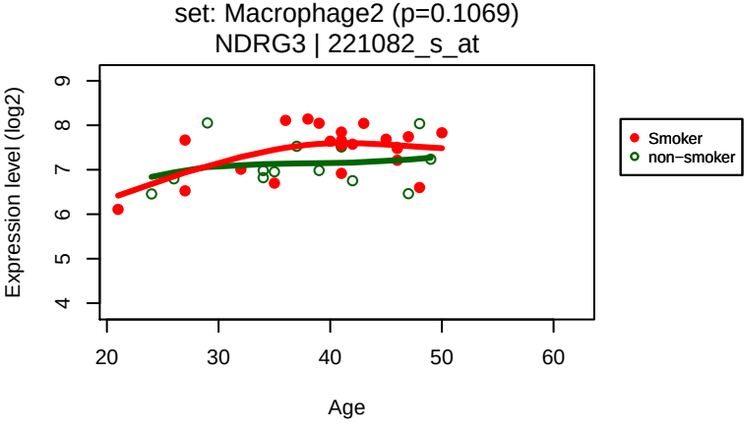

set: SmallAirway2 (p=0.2275)
NDRG3 | 221082_s_at
Expression level (log2)
Age
Smoker
non−smoker

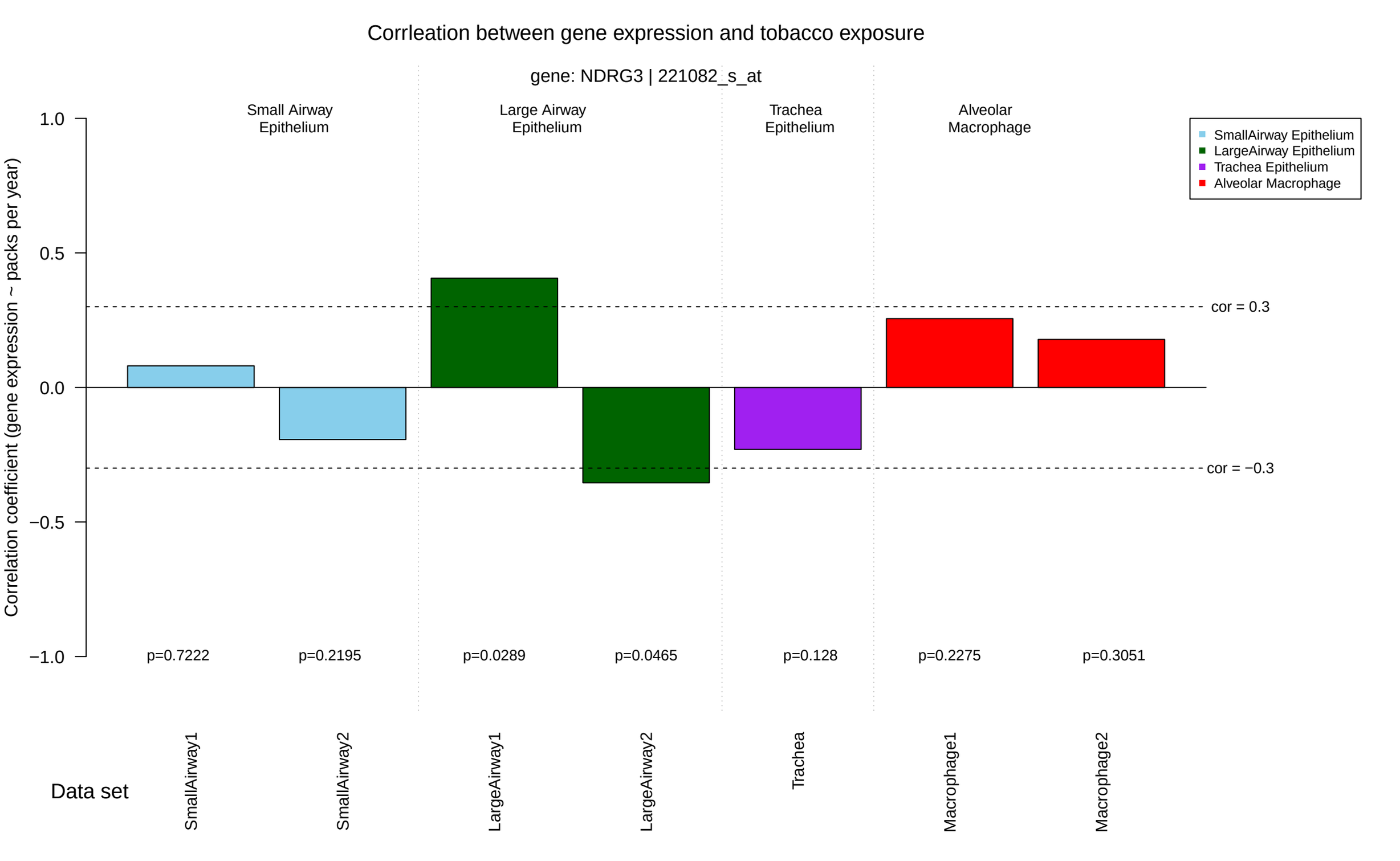

set: LargeAirway1 (p=0.3452)
NDRG3 | 221082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0127)
NDRG3 | 221082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0385)
NDRG3 | 221082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.684)
NDRG3 | 221082_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1069)
NDRG3 | 221082_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NDRG3 | 221082_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7222
p=0.2195
p=0.0289
p=0.0465
p=0.128
p=0.2275
p=0.3051
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2